Variability and acceleration in the growth of children's early vocabulary

Michael C. Frank

Stanford University

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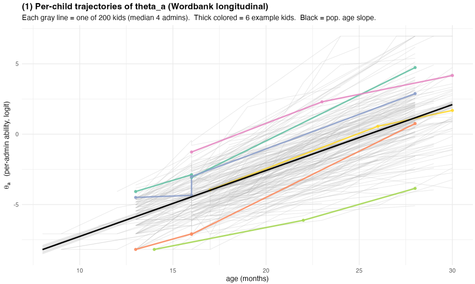
Vocabulary growth: rapid, variable, consequential

Theoretical: Growth mechanisms are diagnostic of how children's learning works — and how it changes with development.

Practical: Variation in vocabulary growth is a key precursor of academic outcomes.

Methodological: Vocabulary count has a property nearly no other psychological metric has — a *true zero*. Children at birth know zero words. Absolute variability in word counts is interpretable, not just z-scores.



Per-admin theta on Wordbank longitudinal sample. Each grey line = one child.

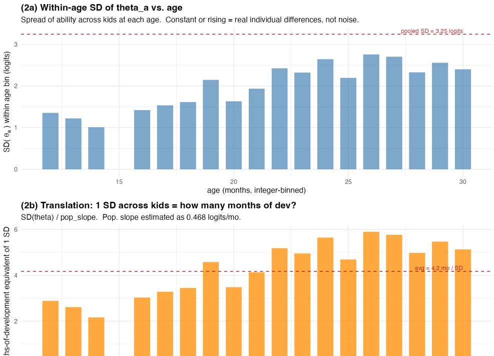
Two signatures of vocabulary growth

What any theory of word learning has to predict

Variability

Children fan out, not converge.

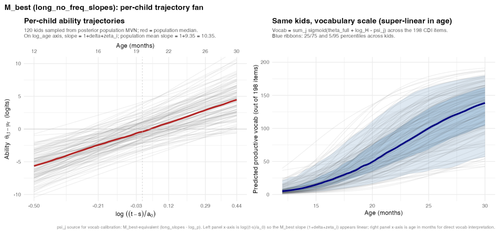
Within-age SD of ability *grows* with age.



Acceleration

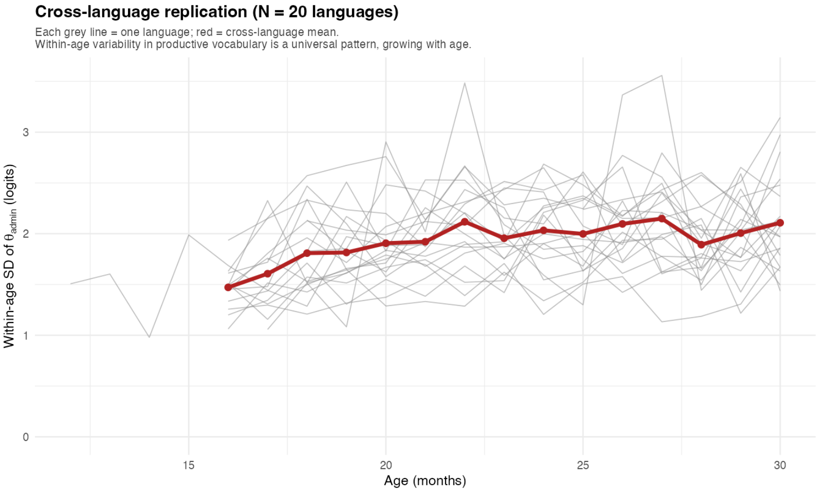
Vocabulary grows super-linearly in tokens.

Population scaling exponent $1 + \delta \gg 1$.



Cross-language replication

Variability and growth-with-age across Wordbank languages



The estimand: a pure accumulator as the structural null

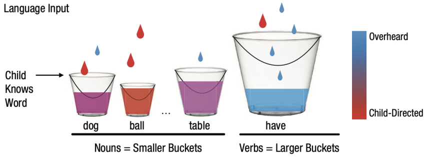
Pure accumulator (McMurray 2007). Each child accumulates word-evidence at a per-child input rate r_i. Word j is learned when threshold ψ_j is crossed.

Predictions.

- Ability $\propto \log(\text{tokens})$, slope $= 1$.
- Between-child variance proportional to σ_r^2.
- No per-child acceleration deviation from population.

We measure deviations from this null. Two estimands carry the structural argument:

δ – *population age-acceleration*



Bucket-fill metaphor: each child hears tokens at rate $r_i \cdot p_j$; word j is acquired when its bucket fills past threshold ψ_j. Originally Kachergis et al. 2021.

Linear predictor

Item-response theory + accumulator dynamics

$$P(\text{produces}_{ijt} = 1) = \sigma\big(\eta_{ijt}\big)$$

$$\eta_{ijt} = \underbrace{\xi_i}_{\text{child}} + \underbrace{(1 + \delta + \zeta_i)}_{\text{age dynamics}} \log \tfrac{t-s}{a_0} \;-\; \underbrace{\psi_j}_{\text{word}}$$

Child intercept $\xi_i = \log r_i + \log \alpha_i$.

- $\log r_i$: input rate, externally pinned via Sperry / HR / WF.
- $\log \alpha_i$: efficiency multiplier (free).

Acceleration $1 + \delta + \zeta_i$: scaling exponent on log-tokens. Population mean δ, per-child deviation ζ_i.

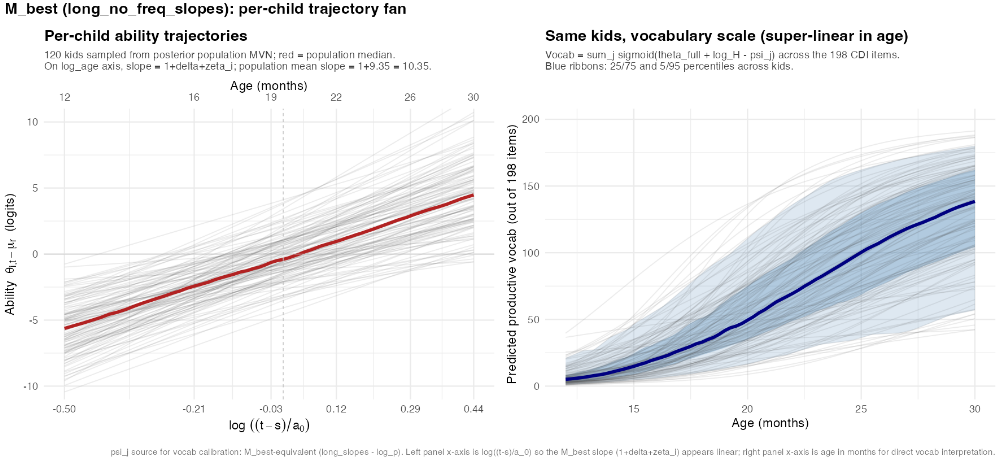
Word threshold ψ_j: log-tokens needed to acquire word j, hierarchical by lexical class.

Headline derived quantities:

- $\pi_\alpha = \sigma^2_\alpha / (\sigma^2_\alpha + \sigma^2_r)$: efficiency-share of intercept variance.
- σ_ζ: between-child growth-rate variance.

M_best fit: per-child trajectory fan

120 kids drawn from posterior of `long_no_freq_slopes`



Left: ability $\theta_{it} - \mu_r$ vs. $\log_{age}$, slope $= 1 + \delta + \zeta_i$, mean ≈ 10.4. **Right:** same kids,

M_best parameters

Variability and acceleration both bounded firmly above zero

Parameter	Posterior	What it does
σ_α	1.71 [1.55, 1.91]	Between-child efficiency variation.
$\boldsymbol{\pi_\alpha}$	**0.91 [0.89, 0.93]**	**Share of intercept variance from efficiency, not input.**
$\boldsymbol{\delta}$	**9.35 [9.21, 9.51]**	**Population age-acceleration: ability scales as $T^{1+\delta}$.**
$\boldsymbol{\sigma_\zeta}$	**3.47 [3.14, 3.84]**	**Between-child variation in growth rate.**
$\rho(\xi, \zeta)$	$+0.35$ $[+0.22, +0.47]$	Higher-baseline kids have steeper growth.

Variability

$\pi_\alpha = 0.91$: efficiency variance dominates input variance.

Acceleration

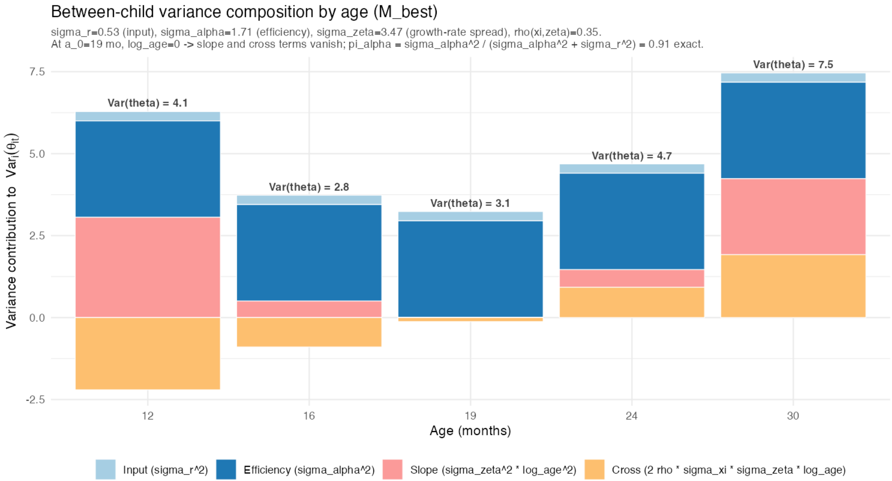
$\delta = 9.4$: scaling is wildly super-linear, far from McMurray's $\delta = 0$.

Growth-rate variance

$\sigma_\zeta > \sigma_\alpha$: kids vary more in growth rate than in baseline.

Where between-child variance comes from, by age

π_α is exact at a_0 and qualified elsewhere



Cross-sample variability: π_α replicates

Five samples, two languages, three input-observation channels

Sample	N	σ_α	π_α
English Wordbank longitudinal	200	1.83 [1.65, 2.04]	0.91 [0.90, 0.93]
Norwegian Wordbank longitudinal	200	2.10 [1.90, 2.35]	0.94 [0.93, 0.95]
BabyView (head-mounted video)	20	1.13 [0.86, 1.61]	0.84 [0.68, 0.93]
SEEDLingS (LENA + comp correction)	44	1.39 [1.13, 1.74]	0.94 [0.89, 0.97]
Peekbank-Stanford (LWL processing)	62	1.64 [1.34, 2.03]	0.90 [0.86, 0.94]

Headline. $\pi_\alpha \in [0.84, 0.94]$ across every well-conditioned sample.
Input rate quantity explains $<10\%$ of between-child intercept variance.

Cross-channel replication: directly observed input (BabyView, SEEDLingS), inferred input (Wordbank), processing-coupled (Peekbank). SEEDLingS extreme corrected for parent-report noise via comprehension channel.

Cross-language acceleration: where it lives, not whether it does

Sample	**admins/kid**	$\boldsymbol{\delta}$	$\boldsymbol{\sigma_\zeta}$	**mean of** $1+\delta+\zeta_i$
English	median 3	9.4	3.5	$\sim$10.4
Norwegian	median 8	11.5	3.7	$\sim$12.5

Population scaling exponent $\sim 10-12$ **in both languages.**

What differs across languages is the *decomposition*. With dense longitudinal data (Norwegian: 8 admins/kid), per-child slopes ζ_i are well-identified, and population δ alone — without ζ_i — actually *hurts* the fit (-235 ELPD).

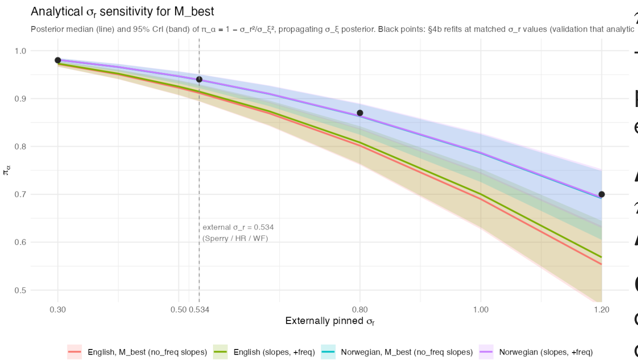
With sparse data (English: 3 admins/kid), ζ_i is hard to pin down, and δ pools the action.

Reading. The acceleration signature is universal. The parameterization that captures it is identifiability-dependent on data density.

Concretely: in any sufficiently dense longitudinal sample, between-child variation in $1+\delta+\zeta_i$ *spans roughly* $5-15$.

Stress test 1: how robust is π_α to the input-rate prior?

Analytical sensitivity, validated by refits



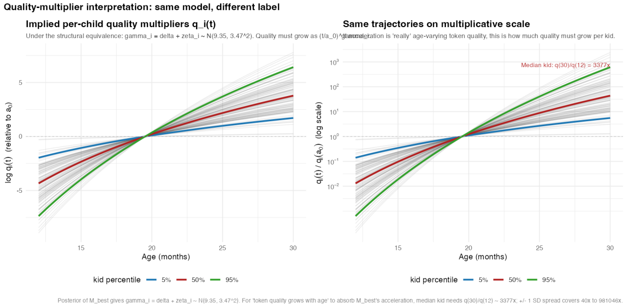
$\pi_\alpha(\sigma_r) = 1 - \sigma_r^2/\sigma_\xi^2$.

The data identifies σ_ξ^2 tightly; the σ_r prior just *partitions* it into input vs. efficiency.

At external $\sigma_r = 0.534$: $\pi_\alpha \approx 0.91\text{–}0.94$.
At doubled $\sigma_r = 1.2$: π_α still > 0.55.

Conclusion. Efficiency-dominated decomposition holds across any defensible external σ_r.

Stress test 2: can input quality absorb acceleration?



Reviewer reflex. "Maybe kids' tokens differ in *quality*, and that's what looks like acceleration."

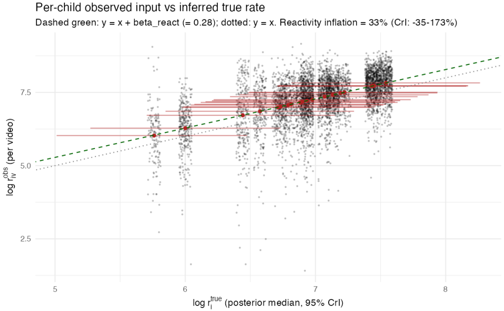
Structural answer. Per-child age-varying quality $q_i(t) = (t/a_0)^{\gamma_i}$ on input is observationally equivalent to $\gamma_i = \delta + \zeta_i$.

Implication. Quality cannot escape acceleration — it *relabels* it. For the median kid, "quality" would have to grow $\sim$ 3,000$\times$ from age 12 to 30 mo. For the 95th percentile kid, $\sim 10^6\times$. Implausible as a substantive quality story.

Replicating with directly observed input

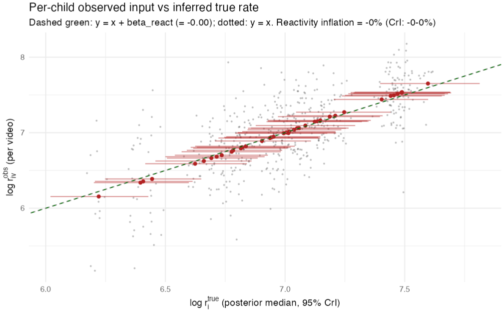
BabyView (head-mounted video) and SEEDLingS (LENA all-day audio)

BabyView, $N = 20$



$$\pi_\alpha = 0.84\ [0.68, 0.93]$$

SEEDLingS, $N = 44$ (with comp correction)



$$\pi_\alpha = 0.94\ [0.89, 0.97]$$

With per-recording observed input rates, the π_α headline holds. Input-rate variance is a small

Adding a processing-speed readout: Peekbank LWL channel

$N = 62$ Stanford-linked subjects with both CDI and looking-while-listening

Joint model adds a per-child LWL log-RT readout coupled to $\log \alpha_i$ via γ_{rt}.

Posterior.

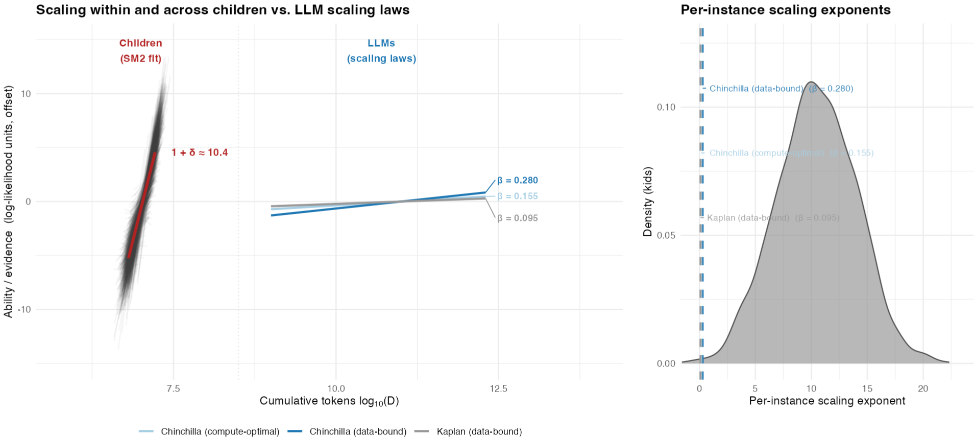
- $\gamma_{rt} = 0.082\ [0.046, 0.125]$ — LWL processing speed is real, and bounded above zero.
- $\pi_\alpha = 0.90\ [0.86, 0.94]$ — replicates within the Peekbank subset.
- $\rho(\zeta_i, \mathrm{rtslope}_i) = -0.02\ [-0.33, +0.27]$ — CDI-side and LWL-side growth rates do *not* share variance.

Two clocks, not one. The acceleration in vocab growth and the maturation of processing speed are independent maturation processes within the same child — different "efficiency clocks" ticking in parallel.

This rules out the simplest unification ("one underlying efficiency axis"); the two readouts each pick up a different developmental process.

Coda: how does this compare to LLM scaling laws?

Slopes are the structurally meaningful comparison; y-axis offset is anchored for visual clarity. Kid D range comes from per-kid r_i.



SM2 fit 'long_slopes': median $1+\delta = 10.39$ [10.24, 10.55]; $\sigma\_\zeta = 3.48$ [3.15, 3.87]; mu_r=7.34, sigma_r=0.53, a_0=19. Chinchilla and Kaplan exponents from Hoffmann et al. 2022 / Kaplan et al. 2020.

Mechanisms not in the standard model

What human learning needs that pure accumulators don't have

Sources of variability

- Interactional input quality (joint attention, contingent talk)
- Genetic / heritable variation
- Environmental richness not captured by token quantity
- Processing-side individual differences (working memory, attention)

Sources of acceleration

- Growing processing efficiency with maturation
- Learning-to-learn (mutual exclusivity, shape bias, syntactic bootstrapping)
- Domain-general changes in attention, memory, perception
- Cross-word leverage / vocabulary-mediated bootstrapping

The structural finding constrains theory: any account of early word learning must produce both super-linear-in-tokens growth and large per-child variation in growth rate.

Summary

- Vocabulary uniquely affords *absolute* variability measurement — there is a true zero, and word counts are interpretable in word-count units.
- Two robust signatures across $\sim$30 languages and 5 modeling samples: **variability** ($\pi_\alpha \approx 0.91$, growing with age) and **acceleration** ($1 + \delta + \zeta_i \sim 10$, varying widely across kids).
- Both signatures violate a pure-accumulator null. Posited "quality" alternatives *relabel* acceleration rather than escape it.
- The picture is structurally different from LLM scaling laws on both axes (steeper exponent, between-instance variance).
- Open question: *which* mechanism — maturation, learning-to-learn, processing speed — is the load-bearing source of the acceleration we measure?

`github.com/langcog/standard-model-2`